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COMPARISON OF PRESSURE SWING METHOD VS ENTRAINER METHOD FOR PRODUCTION OF ANHYDROUS ETHANOL USING ASPEN HYSYS

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Abstract

This study explores the azeotropic distillation of ethanol-water azeotropic mixture using two separation techniques: Pressure Swing Distillation (PSD) and Extractive Distillation (ED) using cyclohexane as an entrainer. Given the limitations of conventional distillation in separating azeotropes, the two distillation methods were simulated with Aspen HYSYS to evaluate their economic performances. The focus of the analyses was on total cost that include capital cost, operating costs, utility cost, and additional materials. From the simulation results, despite both methods having the same operating condition, the PSD method was shown to have lower total cost associated with raw materials, operating, and utility costs than ED method. The greater cost was a result of the additional solvent (cyclohexane) and the recovery processes associated with it, although initial capital cost for the ED method was similar to the PSD method.

This economic analysis indicates that under the operating conditions of this simulation, Pressure Swing Distillation can produce more economically viable separations for ethanol-water separation, making it favorable for applications where cost minimization is a priority.

1-Introduction

Ethanol is an extensively used chemical that has great industrial significance in all areas, it is used in pharmaceuticals, cosmetics, food processing, and increasingly as a renewable fuel such as bioethanol. The push for cleaner energy sources worldwide has increased the demand for high purity ethanol with applications typically requiring percentages greater than 99%. A

separation with this level of purity has a major separation problem because of the

azeotrope properties of the ethanol-water mixture.

The azeotropic properties for ethanol-water solutions show they form a minimum-boiling azeotrope at approximately 95.6% ethanol by weight and past this point traditional distillation is not an effective means of separation.

Because traditional distillation will not work past this concentration, other sophisticated separation technologies, such as azeotropic distillation and pressure-swing distillation, can be applied to provide ethanol at desired purity

Azeotropic distillation is a method of separation in which an entrainer is added to the mixture to change the relative volatilities of the components of the mixture. If there is an interaction between the entrainer and the original components that is effective, it is possible to break the azeotrope and achieve separation. Azeotropic distillation can be homogeneous (single-phase) or heterogeneous (multi-phase) based on the interaction of the entrainer. Heterogeneous azeotropic distillation is appealing to manufacturing because of its low energy consumption, ease of entrainer recovery, and simplicity of operation. An alternative technique, pressure-swing distillation (PSD), takes advantage of the fact that the azeotropic composition of the ethanol-water mixture varies with pressure. By employing two distillation columns operating at different pressures, the process enables the separation of ethanol without the need for entrainers, making it environmentally and operationally favorable in certain scenarios.

The study uses a process simulation software called Aspen HYSYS, to evaluate the different methods of separating ethanol from water in both extractive distillation method and pressure-swing distillation method. The simulation evaluates the reflux ratio, feed stage, design of column, the distillation methods in terms of improvement and comparison of

thermodynamic performance parameters and design parameters. There will be an economic analysis and comparison for each method to evaluate the feasibility of each method at a larger-scale processing for ethanol purification. The results will be used to suggest optimal design and operation of distillation systems and to produce ethanol purity greater than 99%. [1]

2-Literature review

The distillation process is carried out by feeding a mixture of liquids to the column at a certain location, called the feed point. The boiler is used to heat the mixture, the boiler is located at the bottom of the column, to form vapor. The liquid flows down through the column, while the vapor rises up. The vapor is cooled in the condenser, located at the top of the column, to form a liquid. The condensed liquid, called the distillate. An azeotropic mixture is a liquid mixture of definite composition that has the same boiling temperature. When the vapor of the azeotropic mixture is condensed, the resulting liquid mixture has the same composition. In other words, the composition of this mixture cannot be changed by simple distillation. the azeotropic mixture form when the vapor-liquid equilibrium curve (VLE) crosses the 45-degree diagonal line on a phase diagram. The composition of the vapor phase is the same as the composition of the liquid phase at the azeotropic point [2]

Types of Azeotropes

Azeotropes can be characterized based on their boiling behavior and phase behavior:

Based on Boiling Behavior:

-**Minimum boiling azeotropes**: The mixture's total vapor pressure exceeds that of any component, leading to a lower boiling point which creates a boiling point that is lower than what would be expected. This is due to low intermolecular forces increasing volatility.

Around 90% of known azeotropes are of this type.

-**Maximum boiling azeotropes**: The mixture's total vapor pressure is lower than any component, resulting in a higher boiling point than either pure component which creates a boiling point that is higher than either pure component. This is due to strong intermolecular forces like hydrogen bonding. These tend to concentrate in the bottoms product during distillation[3]

Based on Phase Behavior:

-**Homogeneous azeotropes**: Exist as a single liquid phase throughout the system. They can be minimum-boiling (like ethanol-water at 101.325 kPa and 78.1°C with 95.6% ethanol by mass) or maximum-boiling. They do not separate into phases upon condensation.

-**Heterogeneous azeotropes**: Exist as two liquid phases due to component immiscibility. This happens when activity coefficients are significantly over 1, causing a component to separate into a distinct phase. Heterogeneous azeotropes always demonstrate minimum-boiling behavior. like (water-n butanol system).

Azeotropic Mixtures Separation Methods

Because conventional distillation is useless in the azeotropic composition point, other methods are used for separation like:

-**Extractive Distillation**: Uses a high-boiling solvent to change the liquid phase activity coefficients, enhancing relative volatility. The solvent is added above the feed and exits at the bottom without forming an azeotrope, disrupting the original azeotrope.

-**Salt Distillation**: A variation using a soluble ionic salt dissolved in the upper reflux to alter relative volatility. The non-volatile salt stays in the liquid phase.

-**Pressure-Swing Distillation**: two columns operating at different pressures to separate pressure-sensitive azeotropes. This technique exploits the change in relative volatility with pressure. For minimum-boiling azeotropes like ethanol-water, products are collected from the bottom streams. It allows for potential heat integration between the high-pressure and low-pressure columns.

-**Azeotropic Distillation**: Involves adding a third component (entrainer).

-**Reactive Distillation**: Introduces a separation agent that selectively and reversibly reacts with a component; the reaction product is distilled, and the reaction reversed to recover the original component

3-Simulation and Methodology

Aspen HYSYS was used to simulate the steady state operation of both PSD and ED methods. This software is known for modeling chemical processes using thermodynamic and energy balance calculations.

Aspen HYSYS allowed evaluation of process parameters such as pressure - composition- and energy consumption providing a comparative basis for technical and economic performance.[4][5]

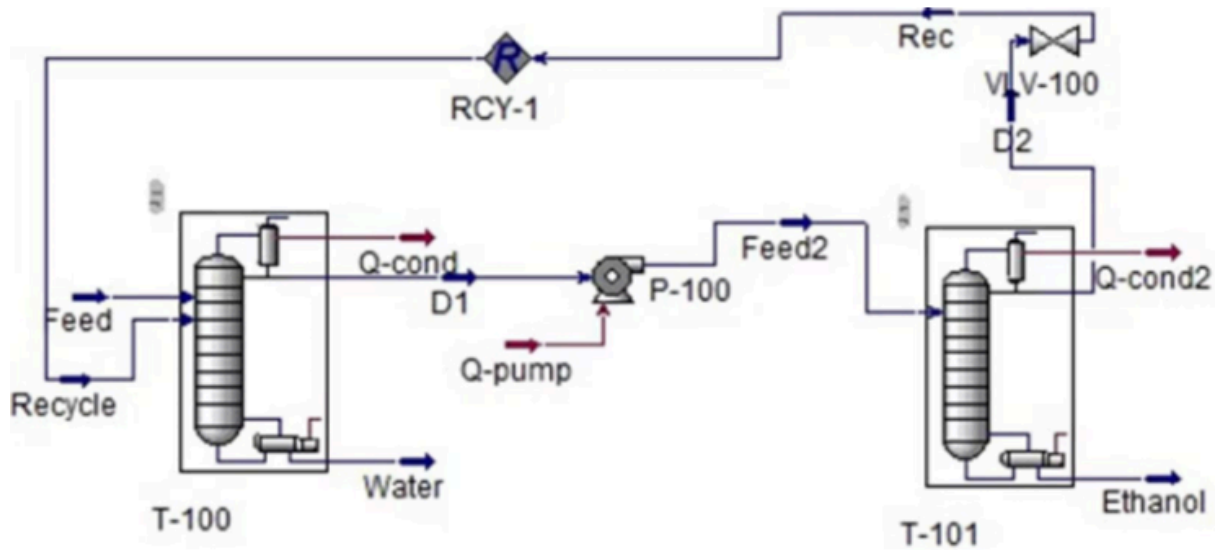


Figure 1: Pressure Swing Method

the PSD method in figure 1 used two distillation columns of different pressures. The low-pressure column (1.2 bar) accomplished a partial separation while the high pressure column (20 bar) allowed additional separation based on a pressure sensitive volatility. Each column was fitted with 30 theoretical stages and the heat integration between the columns was optimized to improve energy efficiency.[6] [7]

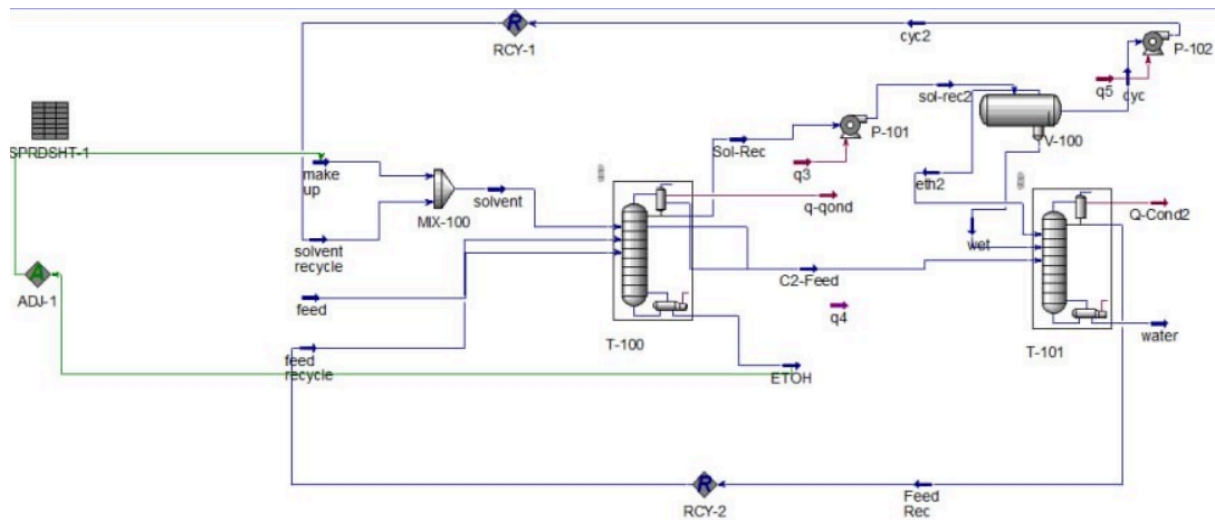


Figure 2: Entrainer Method

With regards to the ED method in figure 2, cyclohexane was used as an entraining agent to form an azeotropic mixture with ethanol and water. This system also utilized two columns; the first column targeted ethanol and the second column aimed for water purification. Process sustainability improved with the recovery and recycling of cyclohexane using a three-phase separator. T-100 was configured with 62 and T-101 with 50 stages to achieve complete separation. The models were both set with a feed containing an ethanol-water stream of 100

kmol/h with 87 mol-% ethanol and 13 mol-% water. The process configurations were optimized to achieve the set purity levels while minimizing energy consumption.[8][9]

4-Results and Discussion

Simulation outcomes for PSD and ED methods showed clear discrepancies between high ethanol recovery, energy input, and costs. The PSD method was able to produce a final purity of 99% ethanol at a production rate of 53 kmol/h. The ED method achieved a higher purity of 99.95% ethanol in addition to a higher production rate of 86 kmol/h.

Table 1: Cost Analysis

factor	PSD method	ED method	
capital cost	6,583,640	6,346,070	USD
operating cost	3,201,570	4,966,260	USD/year
utility cost	1,518,030	3,256,480	USD/year
total installed cost	3,196,800	2,674,300	USD
Entrainer cost	0	27,007	USD/year
Total cost	14,500,040	17,270,117	USD

Regarding the economics terms in table 1, the capital costs for the two methods were similar, but the operating and utility costs of the PSD method were cheaper. The ED method incurred additional costs for the entrainer (cyclohexane) including operating and utility costs, and the energy requirements for the ED method were higher compared to the PSD method, primarily because of the solvent recovery and having multiple columns configured. The carbon emissions for the ED method were higher, yet the experiment showed the recycling efficiency for the entrainer was better than the PSD method. Overall, the simulation results suggested that using the extractive distillation method has advantages over pressure swing distillation. The distillation of cyclohexane to enhance separation yielded higher product purity and yield, especially during periods near the

azeotropic point of separation; however, the distillation included higher utility consumption and made the operating task more complicated.

Pressure Swing Distillation on the opposite side has the advantage of not requiring entrainers, as well as the complexities and risks of dealing with chemicals as the extractive distillation was highlighted above. The ability to utilize heat integration with columns further helps with energy efficiency. In the end, while Pressure Swing Distillation provided slightly lower values of ethanol purity and product recovery, its overall cost-effectiveness provides numerous qualitative and environmental advantages. The cost-effectiveness and lower carbon footprint may make Pressure Swing Distillation make it attractive for large-scale applications even though high purity may not be required.

5-Conclusion

This study shows that PSD and ED both have the capacity to separate the ethanol-water azeotropic mixtures. PSD offers lower energy usage, lower operating costs and simpler process design which make it favourable for typical industry applications. ED is better suited to provide high quality ethanol at yield for high specification products such as pharmaceuticals and fuel.

The decision to choose PSD or ED ultimately depends on the industry needs, whether that is cost efficiency, sustainability or product purity.

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